

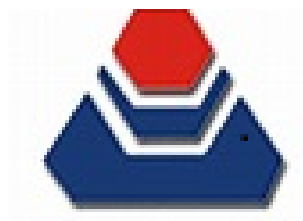
LAB-G :New Approach for Assessment

A project report
submitted in partial fulfillment of the requirements

for the degree of
Bachelor of Technology
in
Information Technology

by
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2022-2023

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Under my guidance and supervision for the award of Degree of Bachelor of Technology in Information Technology

To the best of my/our knowledge and belief the project report

1. Embodies the work of the candidates themselves,
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Date: 25 May, 2023

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The project work as mentioned above is here by being recommended and Forwarded for examination and evaluation.

Date: 25 May , 2023

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Abstract

The most recent Internet innovation has put the entire globe in our hands. Everything proceeds via the Internet, from exchanging information to making purchases. The Internet made the world a small circle. This project is also based on the Internet. This paper shows the importance of chat applications in modern life and their impact on the technological world. The aim of this work is to develop an assessment system for students based on Machine learning and Python.

The application allows a student to take the assessment and get a score by machine learning with only a click. This online platform was created to give students the opportunity to communicate or learn under the direction of professors with the aid of assignments. This application was created with a sound architecture for potential future development.

Insufficient understanding and acceptance of data are crucial challenges while analyzing subjective papers using machine learning. Several attempts have been made to score students' answers using computer science. However, most of the work uses traditional counts or specific words to achieve this task. Furthermore, there is a lack of curated data. This project proposes a novel approach that utilizes various machine learning, natural language processing techniques, and tools such as Wordnet, Word2vec, word mover's distance (WMD), cosine similarity, multinomial naive bayes (MNB), and term frequency-inverse document frequency (TF-IDF) to evaluate descriptive answers automatically. Solution statements and keywords are used to evaluate answers, and a machine learning model is trained to predict the grades of answers. With enough training, the machine learning model could be used as a standalone as well. Experimentation produces an accuracy of 88%.

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Chapter 1

Introduction

1.1 Overview

Lab-G is an application used for assist the growth in lab performance by student. In this application faculty monitor the performance of a student and can see their growth and monitor their lab performance . The app also provides graphical representation and provide prediction of student's performance . Here predication means faculty members check student's learning ability accordingly . We assist student on their assignment performance. It gives students first-hand experience and offers better opportunities for learning. Performing reliable examinations on research material delivered to the labs. Lab Growth is an Idea regarding creating an app which would basically work a student progress report. The automatic evaluation of descriptive question responses using machine learning and natural language processing is proposed in this paper as a new and better method.

A strategy for resolving this issue. The answers are first assessed using the specified keywords and the solution using a variety of similarity-based approaches, such as word mover's distance, and the system compares how similar the modal answer and the student's response are (keeping context in mind) as well as whether or not any keywords are present. Therefore, a student's response that "Karachi is the capital of Pakistan and it is known for its mountain scenery" might receive 50of the marks, "Islamabad and mountain scenery" might receive 30% of the marks because the main keywords are present even though the context is absent, and "Islamabad is the capital and its known for mountain scenery" might receive 100% of the marks because it satisfies all the criteria. Having said that, there are still a number of crucial distinctions between

subjective and objective solutions. They are longer than the objective questions, for starters. Second, writing them requires more time. Additionally, they require a lot more focus and neutrality from the teacher grading them due to the fact that they contain a lot more context. It is challenging to evaluate these problems using computers, mostly because natural language is confusing. Before dealing with the data, a number of preprocessing procedures must be carried out, including data cleansing and tokenization. Then, you may compare the textual data using a variety of methods, including document similarity, latent semantic structures, concept graphs, and ontologies.

The final score can be assessed based on language, structure, and similarity. Due of its one defining characteristic, context, subjective tests are seen by both students and teachers as being more complicated and intimidating. In order to score a subjective answer, the checker must carefully read every word. The overall score is greatly influenced by the checker's mental state, level of weariness, and objectivity. Therefore, letting a system handle this time-consuming and occasionally crucial duty of analysing subjective responses is significantly more time and resource-efficient.

Machine evaluation of objective responses is relatively simple and practical. A programme that can swiftly map students' responses can be fed questions and one-word responses. But dealing with subjective responses is far more difficult. They have a wide range of lengths and a huge vocabulary. In addition, people typically convenient abbreviations, which makes the process that much tricky. Much work has been done on the topic of subjective answers evaluation in one form or another, such as measuring Similarity between different texts, words, and even documents, finding the context behind the text and mapping it with the solution's context, counting the noun-phrase in the documents, matching keywords in the answers.

1.1.1 History

Many genius individuals contributed to its development. But there's one person who stands out when thinking about when ML was invented. Arthur Samuel, a computer scientist at IBM and a pioneer in AI and computer gaming, coined the term "Machine Learning" in 1952." Machine learning is popular because computation is abundant and cheap Machine learning algorithms automatically build a mathematical model using sample data – also known as "training data" – to make decisions without being specifically programmed to make those decisions.

1.2 Problem Statement

1.2.1 Problem in Hand

- Students are not paying attention to Lab lectures.
- Student didn't find an interactive way for showing their knowledge.
- Faculty members are not sure about whether students are getting everything or not.

1.2.2 Project Objective

This kind of automated review significantly improves how the educational sector can carry out its other task and lessens the physical labor required for the mental task ,comparing the answers with the correct answer. Teachers and up use more time training student , creating good curriculum ,or openly checking their result.

1.3 Existing system

- To show the lab performance of the individual student by checking their growth.
- To check the Understandability level of the student.
- It will be helpful for faculty member to assist the student.

1.3.1 Current Scenario

- We don't have the statistical data for the student lab performance.
- Easy to access student data.
- Assessment reailbilty.
- Solution for all student data storage of score.

Chapter 2

Literature Review

Latent Semantic Indexing approach for evaluating subjective online questions. To create a k-dimensional LSI space matrix, they combined subjective ontologies with Chinese automatic segmentation algorithms. The answers were given as TF-IDF embedding matrices[1], and the term-document matrix was then subjected to Singular Value Decomposition (SVD), resulting in a semantic space of vectors. Synonym and polysemy issues were less of a problem because to LSI[4]. Finally, cosine similarity was used to calculate the degree of similarity between the replies[8]. 35 classes and 850 cases were included in the dataset, and the results revealed a 5% discrepancy between teacher grading and the suggested approach. Kusner et al [7]. presented a novel concept of using Word Use Mover's Distance (WMD) to determine where two texts differ from one another[12]. The vector space constraints were loosened by the system's adoption of a lax WMD methodology and the absence of hyper-parameters. Eight real-world datasets were added, including sentiment data from Twitter and sports stories from the BBC[14]. Two more bespoke models were trained in addition to the Word2vec model from Google News. The testing data was classified using a KNN technique. Therefore, loosened WMD resulted in 2 to 5 times faster classification and decreased error rates.

A method to score short descriptive answers using the lexico-semantic pattern (LSP) was proposed by Kim et al [32]. as a result of its successful use with the morphologically intricate Korean language. To better comprehend the user's intents, LSP might arrange the semantic of the response. Additionally, the keywords were expanded using a synonym list to make them more adaptable to different answer types. Dataset from 88 students was collected and translated to LSP, which was then contrasted with the LSP of the solution to score the response.

As a result, the system performed 0.137 points better than the old method. When evaluating subjective answer evaluation techniques in the past, we looked at their drawbacks. Using the results from our result prediction module to train a machine learning classification model, followed by using the trained model to reinforce the prediction model's results, can result in a fully trained machine learning model, which is what we propose as a new approach to solve this problem in this paper. The result prediction module creates vectors of our data using the word2vec word embedding approach while maintaining the semantic meaning of each word, and then utilises Word Mover's Distance to determine how similar the vectors are to one another[15]. We examine the impact of utilising several machine learning models and compare our results with those obtained using other techniques, such as Cosine Similarity

Chapter 3

SDLC Model

3.1 waterfall Model

Probably the oldest and most well-known model. This model is built like a waterfall, as the name would imply. This model's flow can be compared to a waterfall, which moves from higher to lower points before arriving at its destination. The other step's input is the output of the first step. Up until the software is developed, this process continues. Its simplicity is what distinguishes this model from others.

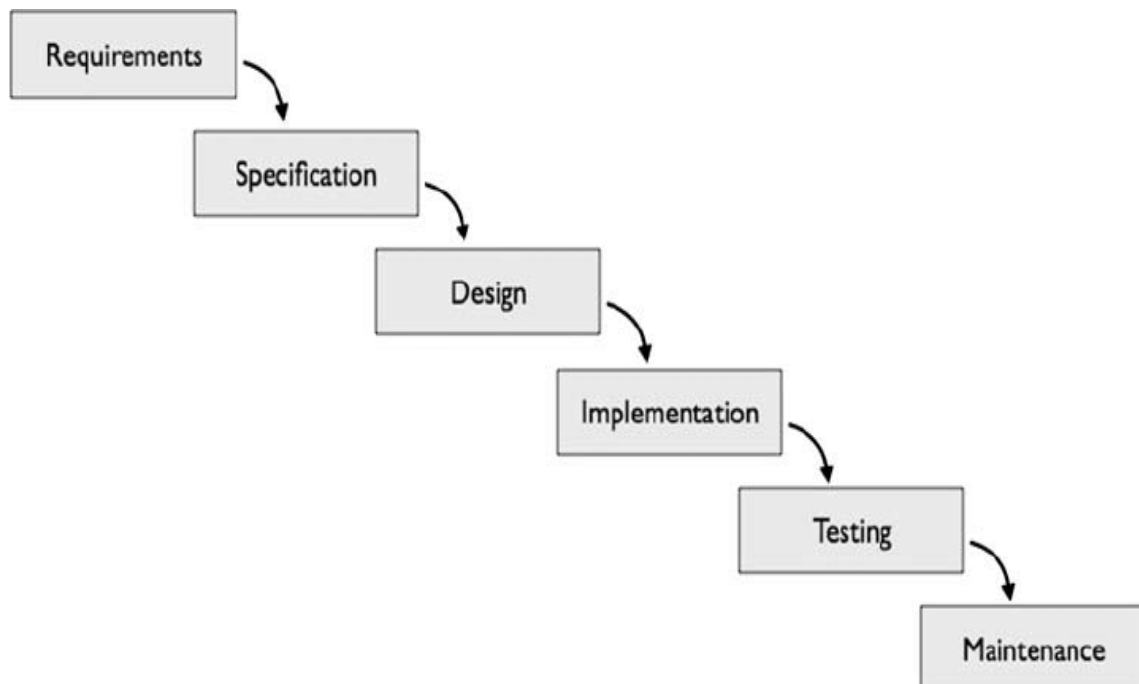


Figure 3.1: waterfall Model

3.1.1 Advantage

- This paradigm is straightforward and easy to understand. Because of this, teamwork is incredibly simple and everyone is on the same page.
- With this model, deadlines are met with ease. This is because the team has previously been given precise instructions and no time is lost trying to figure out how to proceed.
- Since this type of architecture involves a lot of paperwork, it is better suited for larger projects with greater teams.

3.1.2 Disadvantage

- Making adjustments to the product at a later stage in the project is practically difficult, which is one of this model's main drawbacks. This is because the steps are tied to one another in a sequential process, thus changing anything would be exceedingly difficult.
- The inability of the stakeholders and customers to utilise or view the product right away is another drawback of this approach. The customer will have to wait a few months before they can view the goods, which occasionally causes them to feel uncomfortable or disappointed.
- As the product cannot be altered at a later stage of the development process, there is also no space for error. Therefore, extensive investigation is required.

3.1.3 Spiral Model

This model deviates greatly from the norm. It is a hybrid model that combines elements of the waterfall approach and the iterative model. The intriguing feature of this model is that it enables us to evaluate the product at each stage, ensuring the creation of a top-notch final result. This model's software development life cycle differs slightly from the cycles for the other models..



Figure 3.2: Spiral Model

3.1.4 Advantage

- It enables the developers to easily add new functionality to the software that was before impossible to do
- We are able to receive feedback following each spiral, which is a highly advantageous feature that allows us to continuously enhance our offering
- Tasks can be ranked in order of importance based on market demands. We may focus on the iteration that the market demands, for instance, and finish the other iterations later.

3.1.5 Disadvantage

- In order to manage these spirals, highly qualified managerial personnel are needed. If management is poor, there is a chance that the spiral will continue forever.
- Due to the complexity of this process, there is an increased risk of error.

3.2 Proposed Model

3.2.1 Iterative Waterfall Model

We are using Iterative Waterfall Model :

- The Iterative Waterfall Model is a method of software development that combines iterative design flexibility with the sequential processes of the conventional Waterfall Model.
- The primary distinction between the iterative waterfall model and the traditional waterfall model is the provision of feedback channels from each step to its predecessor phases.
- It is a good choice for small projects or teams who are new to software development because it is simple to comprehend and put into practice.

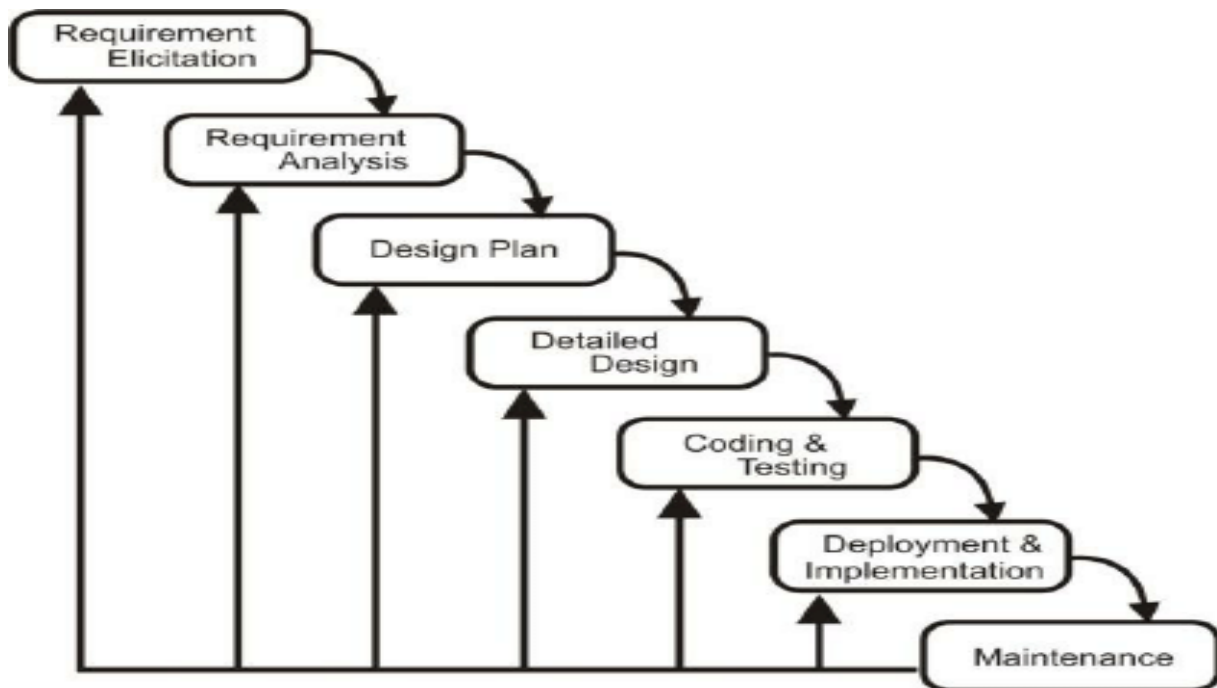


Figure 3.3: SDLC- Iterative Model

In this model, the software development process is divided into a series of iterations, or cycles, each of which involves the following phases:

Feasibility Study

Feasibility is the degree to which a project can actually be carried out successfully. A feasibility study is conducted to assess the solution's viability, which establishes whether it is viable and implementable in the programme. The objective of the feasibility study is :

- To ascertain whether the programme can be implemented using the available technology and within the budget and time constraints
- To determine whether the software can be integrated with other existing software

Requirement Analysis and Specification

The various project requirements, including the required software and hardware, databases, and interfaces, are determined in this stage of the waterfall paradigm. The Kaggle emotion recognition database served as the dataset, Python was utilised as the programming language, frontal faces of various objects were collected to offer the feature extraction data, and Twilio was used as the cloud communication platform.

Design Phase

During the project's implementation phase, we successfully implemented the various modules needed to achieve the desired results at the various module levels. The modules created for facial detection, text mining, and user communication are involved.

Coding Phase

During the project's implementation phase, we successfully implemented the various modules needed to achieve the desired results at the various module levels. The modules created for facial detection, text mining, and user communication are involved. The system is initially built in tiny programmes known as units with input from the system design, and is then combined in the following phase. Unit testing is the process of developing and evaluating each unit for functionality.

Testing Phase

To determine whether the project module is delivering the anticipated outcomes within the estimated timeframe, the numerous test cases are executed. The entire system is merged after each item made during the implementation phase has been tested. After integration, the entire system is checked for flaws and defects.

Deployment

The product is either installed in the customer environment or made available for purchase following functional and non-functional testing.

Maintenance

When a product has undergone both functional and non-functional testing, it is either put into use in the environment of the customer or made available for purchase.

Chapter 4

Feasibility Study

Problem Statement Feasibility Assessment Using, Satisfiability Analysis and Np Hard, Np-Complete or P Type Using Modern Algebra and Relevant Mathematical Models

4.1 NP-Analysis Of System

- **NP Complete :** When a problem can be solved in polynomial time using non-deterministic techniques, it is said to be NP-Complete. A judgement question is said to be in NP if an exponent k exists and a non-deterministic method for it runs in time $O(n^k)$ for all hints, where n is the length of the input. The polynomial time required for feature extraction to obtain new variables from the image's matrix in order to construct unique classes. This makes it NP-Complete. The methods for our project can be determined in polynomial time, however database interaction takes an infinite amount of time. As a result, this project is a database file processing project and is NP-Complete.
- **NP-Hard Complete :** What are NP-Complete and NP-Hard? If a known polynomial-time procedure exists to obtain the solution, the decision issue is in the category P. P is a grouping of all the issues that can be resolved in polynomial time. In other words, a decision question is in P if an algorithm and an exponent k exist. For the query that takes $O(n^k)$ time, where n is the input's length. If there is a known polynomial-time procedure for a non-deterministic machine to obtain the solution, the decision issue is in the NP category. Non-Deterministic Polynomial Problems are those that cannot be solved in finite time or whose complexity of execution cannot be quantified by a mathematical algorithm. A problem is in the class NPC if it is in NP and is as hard as any problem in

NP. A problem is NP-hard if all problems in NP are polynomial time reducible to it, even though it may not be in NP itself.

If a polynomial time algorithm exists for any of these problems, all problems in NP would be polynomial time solvable. These problems are called NP-complete. The phenomenon of NP-completeness is important for both theoretical and practical reasons

4.2 Feasibility Assessment:

Project algorithm can be determined by polynomial time as it has definite and predictable output. This Process is done within a finite amount of time hence belongs to a NP Complete class Problem

Chapter 5

Requirement Analysis and Specification

5.1 System Requirement Specifications (SRS)

We create a android software. It is mostly used to display score and predicted scor and is used to provide information on student and faculty. Utilising the SRS makes it easier to verify that our project's requirements—namely, that the data and assesement feedback. Additionally, it aids in the decision-making process for the project life cycle. Also, creating an SRS might cut down on overall development expenses and time.

5.2 System Overview:

5.2.1 Assumption and Dependencies:

Early on in the life cycle of our project, we made the assumption that the system would be able to predict the scoring of the student with the same accuracy as any faculty done scoring of marks . Answer data should be made accessible for the project at some point during execution. Test data will be used up until that point to carry out the presentations' demo. It is assumed that the user is familiar with using a android deivce. The android studio is required since the application is android-based. Users are expected to have reliable internet connectivity.

5.2.2 Functional Requirement:

- It provides an easy interface to user.

- The accessibility or response time of the application should be fast
- login of student and start the quiz is initial step.
- prediction of score is done only after the assessment of quiz.

The Functional Requirements Specification documents the operations and activities that a system must be able to perform. Functional Requirements should include:

- Dataset must be required.
- Input will be required.

The Functional Requirements Specification is designed to be read by general audience. Readers should understand the system, but no particular technical knowledge should be required to understand the document

System Feature 1 (Functional Requirement)

- A Android is provided as an interface.
- Trained model works efficiently with front end to provide easy working.
- score predicted as soon as quiz is accessed.

System Feature 2 (Functional Requirement)

This section describes the functional requirements of the application and the features it provides. System features are described in detail to help the future extension and testing of the system. Features stated here are already parts of the implemented system so no prioritization is needed. Priority is needed for features to be developed that will be added to this document later.

5.2.3 Non-Functional Requirement:

Performance Requirement

System can produce results faster on 4GB of RAM. It may take more time for peak loads at main node. The system will be available 100% of the time. Once there is a fatal error, the system will provide understandable feed back to the user.

Safety Requirement

- Only administrators have access to the database of each individual user
- All data will be backed-up every day automatically and also the system administrator can back-up the data as a function for him.
- For the safety purpose backup of the database must be required

Security Requirements

- The system is being developed in Python. Python is an interpreted high-level general-purpose programming language and easy to maintain. The system is designed in modules where errors can be detected and fixed easily.
- The user must provide his details and his contact number which will not be disclosed and even no misuse will be done.

5.2.4 Software Quality Attributes

Our software has many quality attribute that are given below:-

- **Adaptability:** This software is adaptable by all users.
- **Availability:** This software is freely available to all users. The availability of the software is easy for everyone.
- **Maintainability:** After the deployment of the project if any error occurs then it can be easily maintained by the software developer.
- **Reliability:** The performance of the software is better which will increase the reliability of the Software
- **User Friendliness:** Since, the software is a android application; the output generated is much user friendly in its behavior.
- **Integrity:** Integrity refers to the extent to which access to software or data by unauthorized persons can be controlled.

- **Security:** Users are authenticated using many security phases so reliable security is provided.
- **Testability:** The software will be tested considering all the aspects.

5.2.5 System Requirements

Database Requirements

- **Logical Database Requirements:** A logical database can stretch over multiple physical hard disks and information files. The data storage unit is still a single database for information retrieval purposes. To have a logical database, all given hard disks and information files must be accessible from a single source.
- **Physical Database Requirements:** A physical database is technically a smaller unit of storage referred to as a company, field, record or table, depending on how much information the physical storage device contains. A field is the smallest unit of storage housing only a single file.

Software Requirements:

Developer Perspective

- **Operating system :** Windows 10 and above
- **Coding Language :** Python 3.8, ANDROID STUDIO , MONGODB ,JAVA
- **IDE :** Python IDLE
- **Editor:** Visual Studio Code
- **Database:** MONGODB

User Perspective

- **Operating system :** Windows 7 and above, Linux, IOS ,ANDROID.
- **Browser:** (Devices that support Android)
- Internet connectivity

Hardware Requirements

Developer perspective

- **Processor:** i5 Processor and above.
- **Hard Disk :** 40GB(min)
- **RAM :** 4GB(min), 8GB Or above (recommended)

User Perspective

- **Processor:** Android Device, i3 Processor and above.
- **Hard Disk :** 40GB(min)
- **RAM:** 4GB(min)

Chapter 6

Design

6.1 System Design/Modelling :

6.1.1 System Architecture

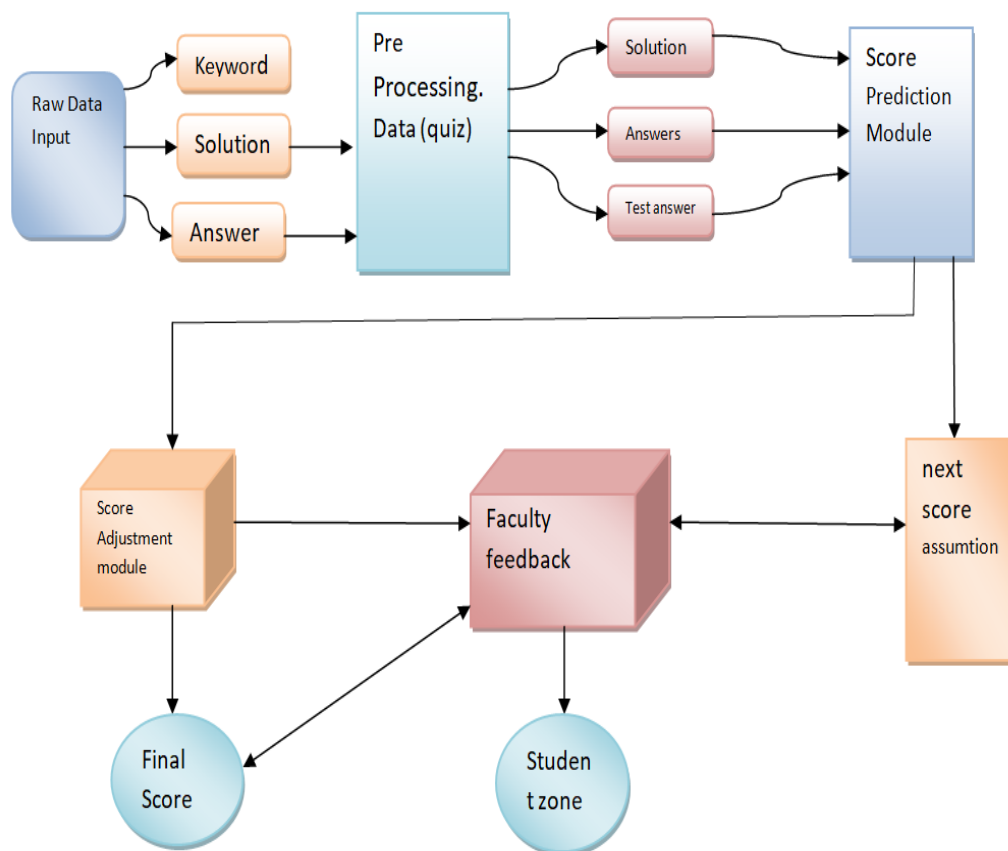


Figure 6.1: Architecture Diagram

6.1.2 Data Flow Diagram

A **Data Flow Diagram (DFD)** is a graphical visualization of the movement of data through an information system. DFDs are one of the three essential components of the structured-systems analysis and design method. A DFD is process centric and depicts 4 main components.

- Processes (circle)
- External Entities (rectangle)
- Data Stores (two horizontal, parallel lines)
- Data Flows (curved or straight line with arrowhead indicating flow direction)

6.1.3 DFD Components

DFD consists of processes, flows, warehouses, and terminators. There are several ways to view these DFD components.[4]

Process

The process (function, transformation) is part of a system that transforms inputs to outputs. The symbol of a process is a circle, an oval, a rectangle or a rectangle with rounded corners (according to the type of notation). The process is named in one word, a short sentence, or a phrase that is clearly to express its essence[2].

Data flow

Data flow (flow, dataflow) shows the transfer of information (sometimes also material) from one part of the system to another. The symbol of the flow is the arrow. The flow should have a name that determines what information (or what material) is being moved. Exceptions are flows where it is clear what information is transferred through the entities that are linked to these flows. Material shifts are modeled in systems that are not merely informative. Flow should only transmit one type of information (material). The arrow shows the flow direction (it can also be bi-directional if the information to/from the entity is logically dependent – e.g. question and answer). Flows link processes, warehouses and terminators.[2]

Warehouse

The warehouse (datastore, data store, file, database) is used to store data for later use. The symbol of the store is two horizontal lines, the other way of view is shown in the DFD Notation. The name of the warehouse is a plural noun (e.g. orders) – it derives from the input

and output streams of the warehouse. The warehouse does not have to be just a data file but can also be, for example, a folder with documents, a filing cabinet, or a set of optical discs. Therefore, viewing the warehouse in a DFD is independent of implementation. The flow from the warehouse usually represents reading of the data stored in the warehouse, and the flow to the warehouse usually expresses data entry or updating (sometimes also deleting data). The warehouse is represented by two parallel lines between which the memory name is located (it can be modeled as a UML buffer node).[2]

Terminator

The Terminator is an external entity that communicates with the system and stands outside of the system. It can be, for example, various organizations (eg a bank), groups of people (e.g. customers), authorities (e.g. a tax office) or a department (e.g. a human-resources department) of the same organization, which does not belong to the model system. The terminator may be another system with which the modeled system communicates.

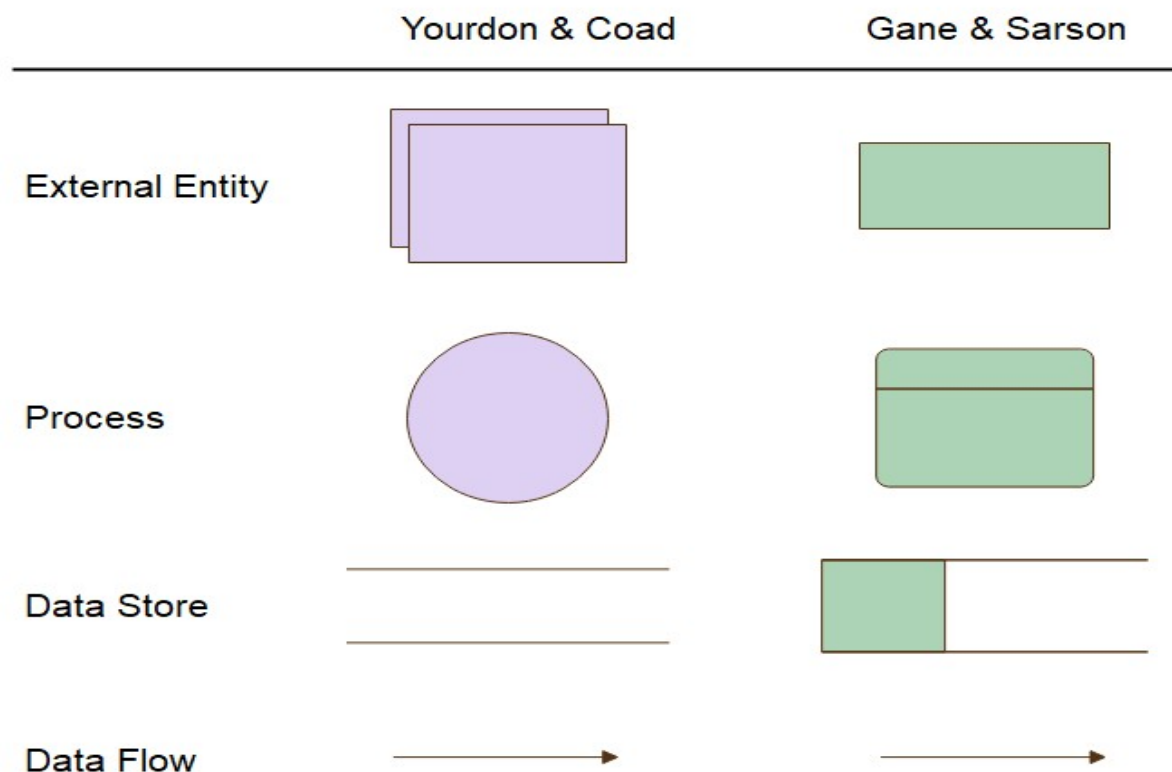


Figure 6.2: DFD Components

0 level DFD

The Context Diagram (0 Level DFD) shows the system under consideration as a single high-level process and then shows the relationship that the system has with other external entities (systems, organizational groups, external data stores, and more). A level 0 DFD is a more thorough diagram than a context diagram and it displays the system's constituent processes. The level 0 DFD will contain a single bubble representing the full software element in the case of LAB-G: student connected to the faculty and their assessment for the quizzes and their assessment, with incoming and outgoing arrows denoting input and output data. The level 0 DFD divides the single bubble into smaller bubbles, which stand in for the many system operations. Until the entire system is drawn out, each bubble on the level 0 DFD will have its own level 1 DFD, which will then be split down into more intricate diagrams. A clear comprehension is provided by the level 0 DFD. understanding of the overall functions and activities of the system

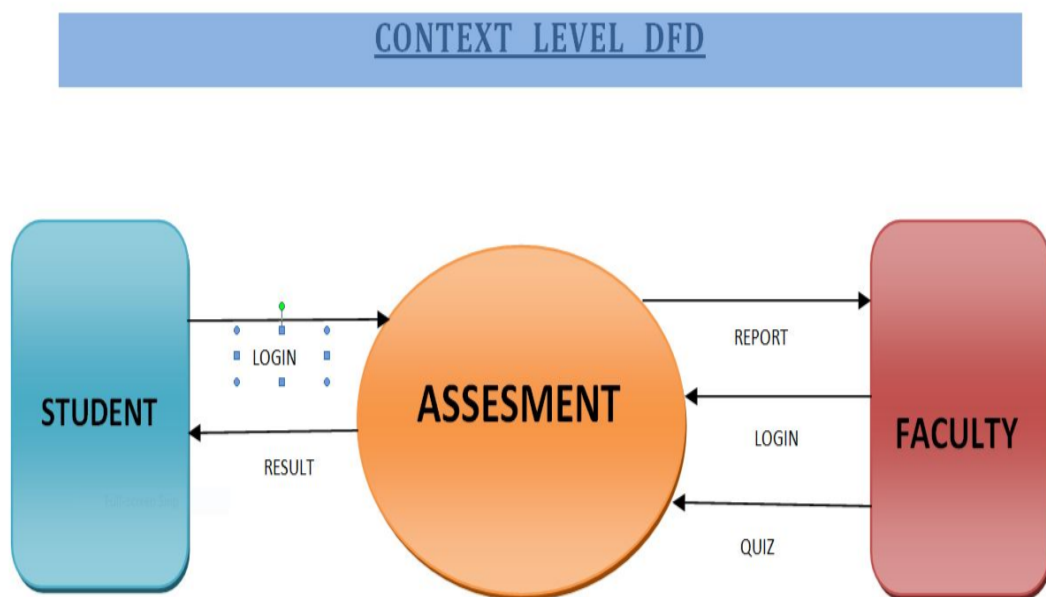


Figure 6.3: Context level Diagram

level-1 DFD

The functional breakdown of the level 0 process is represented by a level 1 DFD of LAB-G: Connecting Students and Faculty Through Assessment in Lab. It deconstructs the processes in greater depth and demonstrates how the processes interact. Multiple processes, each with its own input, output, and data store, are commonly found in a level 1 DFD. A level 1 DFD's processes are identified by numbers, and a brief description of each process' function follows. The level 1 DFD is utilised to get a deeper understanding of the system and serves as the foundation for creating the level 2 DFD, which further deconstructs the level 1 processes.

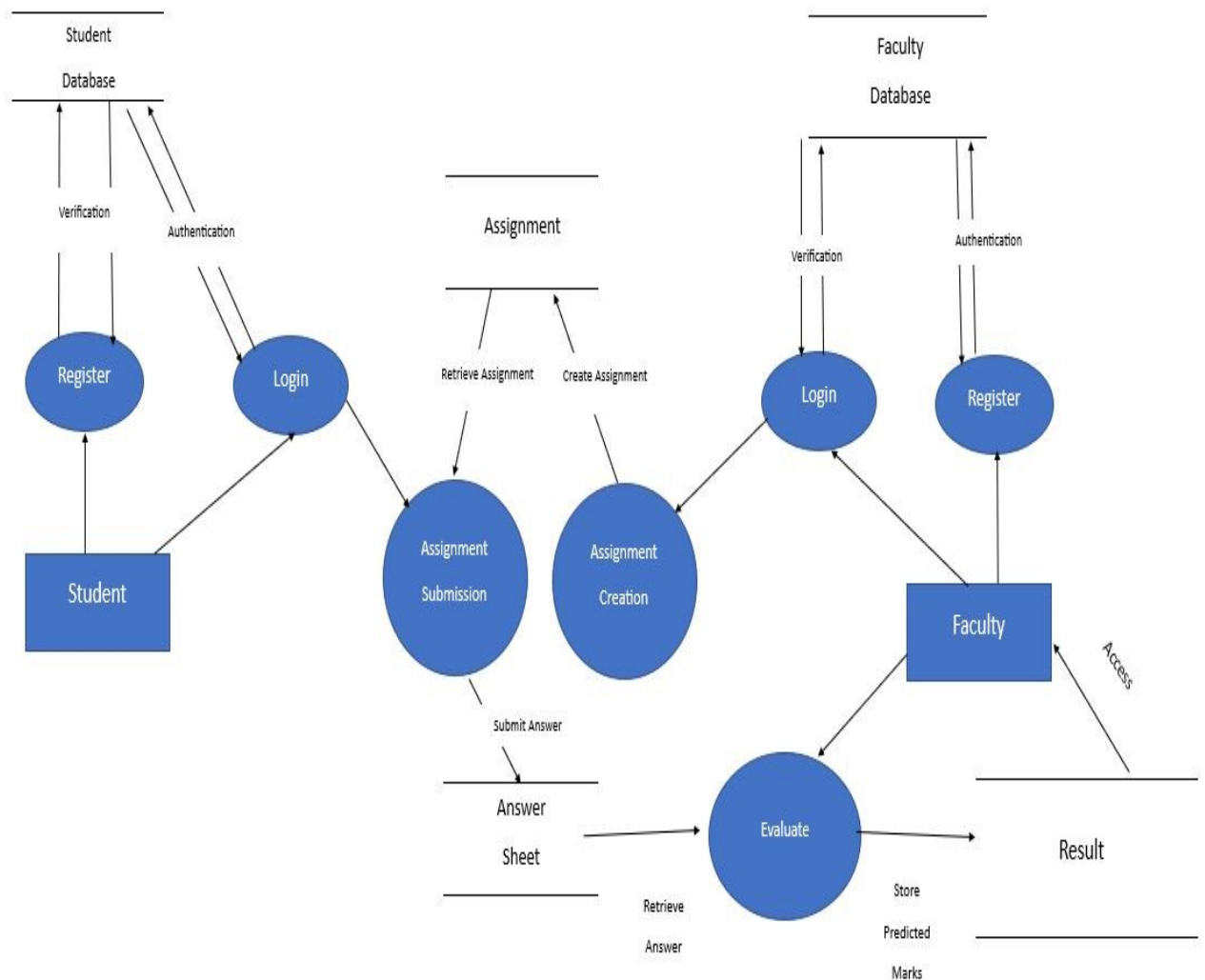


Figure 6.4: level-1 DFD

6.1.4 Use Case Diagram

A use case diagram is a visual representation of the interactions between system components. A process for identifying, outlining, and organising system needs is called a use case. The word "system" here refers to a thing that is being created or run, like a website for mail-order product sales and services.

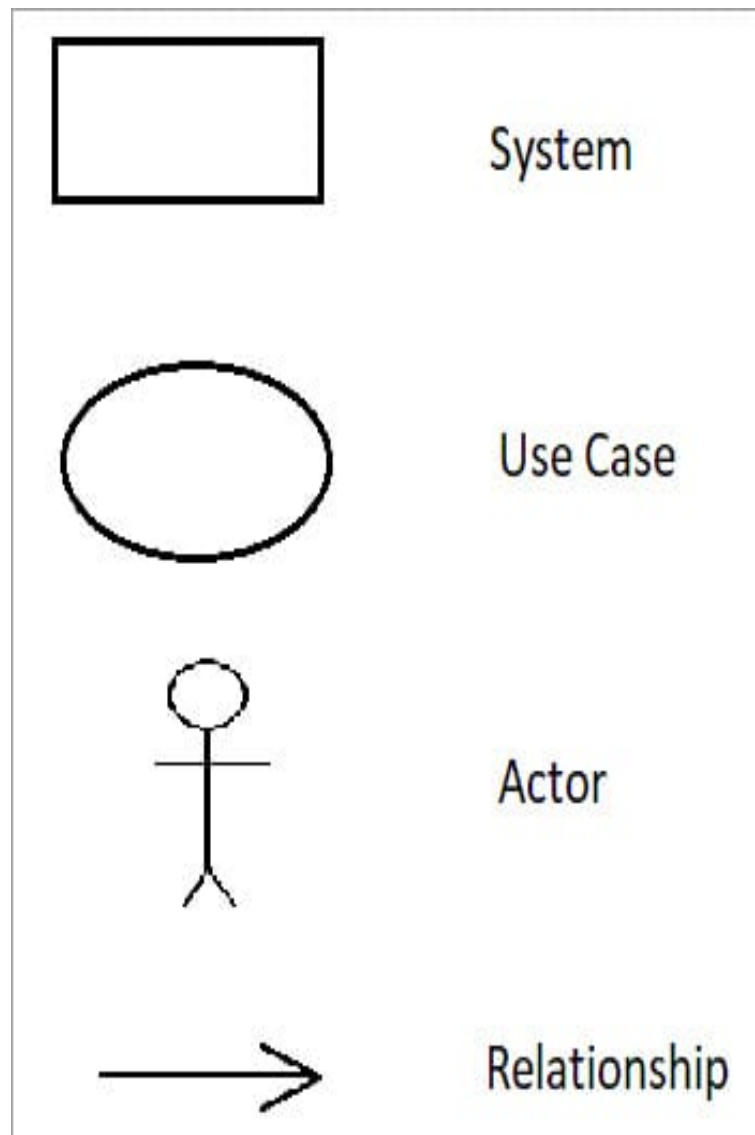


Figure 6.5: Use Case Components

UML (Unified Modelling Language), a standard notation for the modelling of real-world objects and systems, uses use case diagrams.

1. The actors, usually individuals involved with the system defined according to their roles.
2. The use cases, which are the specific roles played by the actors within and around the system.

3. The relationships between and among the actors and the use c.

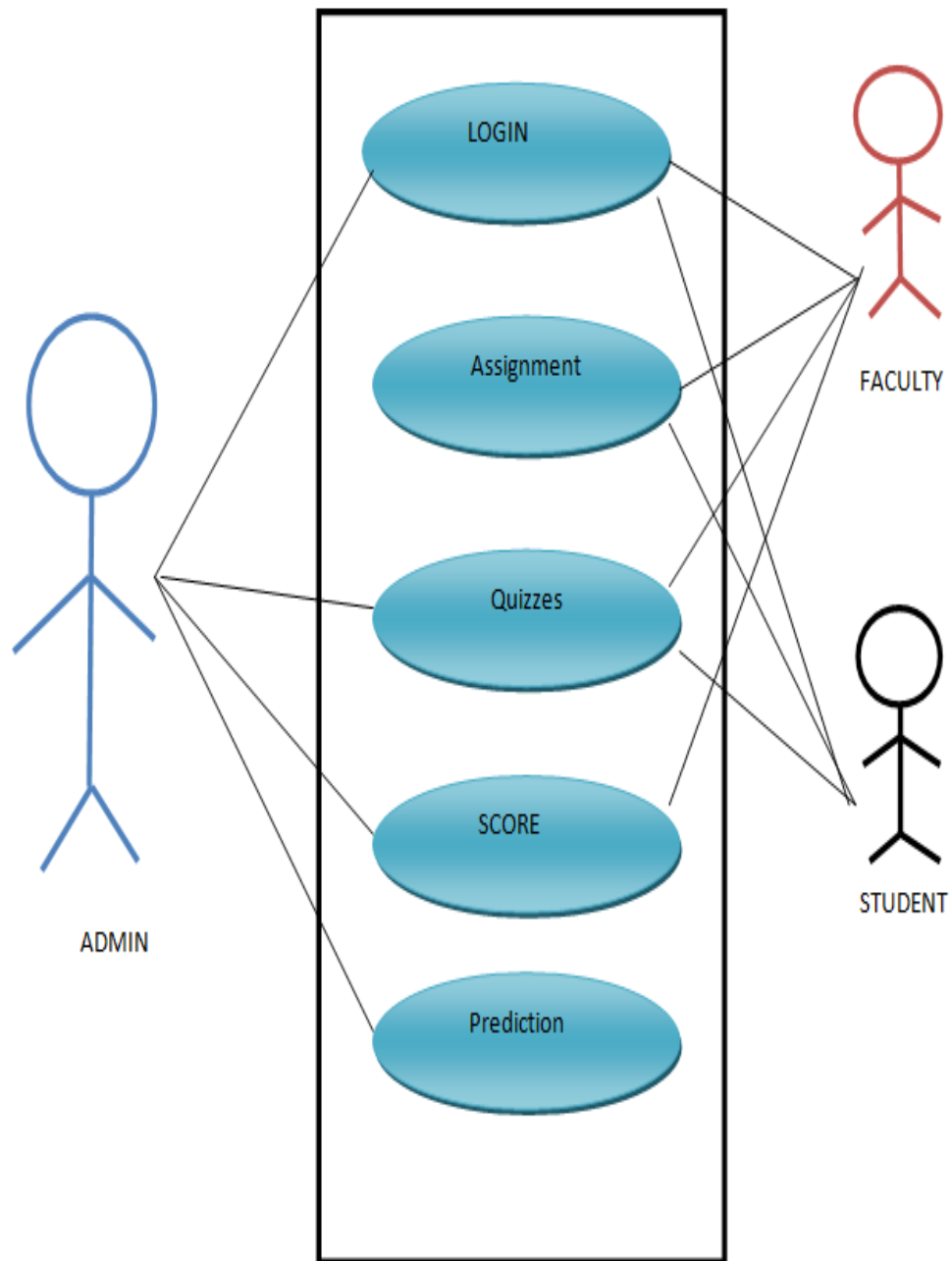


Figure 6.6: Use Case Diagram

6.1.5 Entity Relationship Diagram

An **Entity Relationship diagram** is a high-level data model. This model is used to define the data elements and relationship for a specified system. It develops a conceptual design for the database. It also develops a very simple and easy to design view of data.

ER modeling, it is the database structure which is portrayed as a diagram.

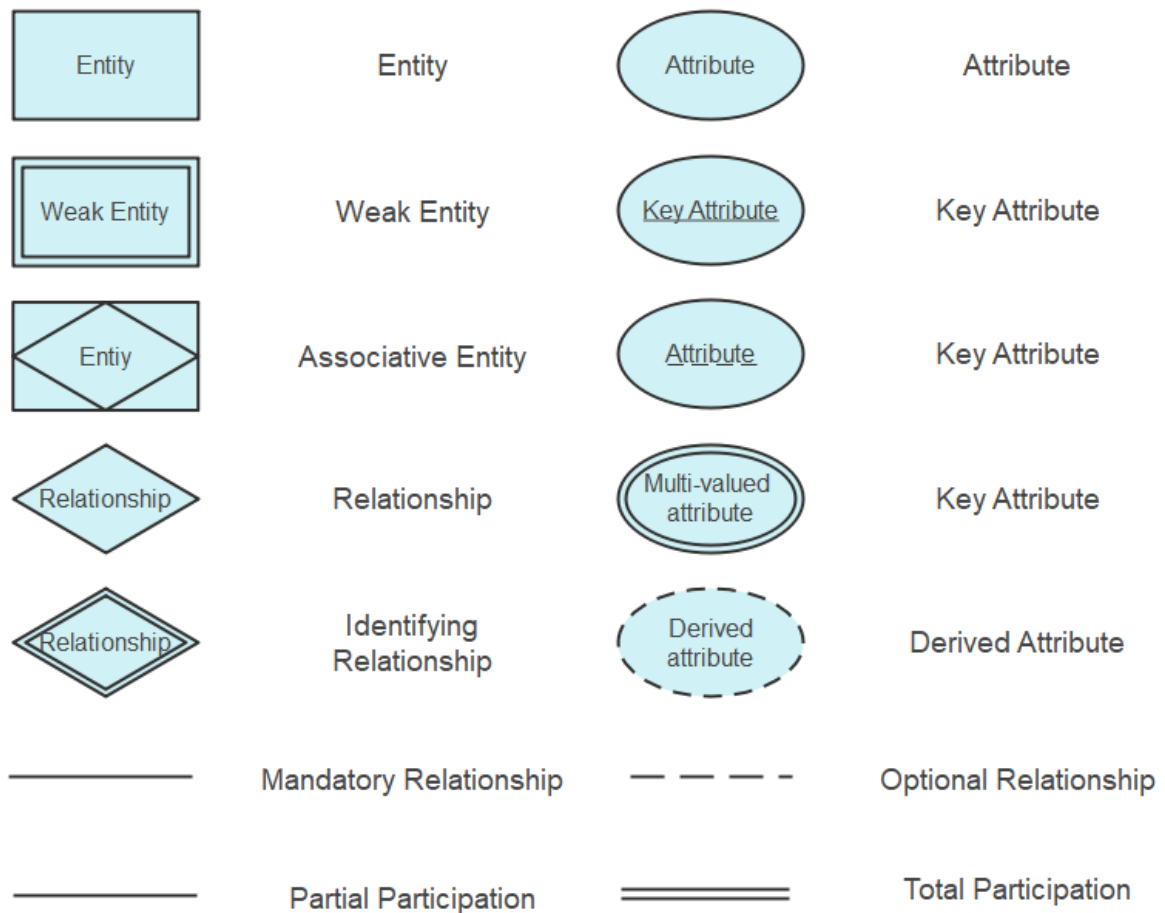


Figure 6.7: Entity Relation Component

ER Diagram stands for Entity Relationship Diagram, also known as ERD is a diagram that displays the relationship of entity sets stored in a database. In other words, ER diagrams help to explain the logical structure of databases. ER diagrams are created based on three basic concepts: entities, attributes and relationships. ER Diagrams contain different symbols that use rectangles to represent entities, ovals to define attributes and diamond shapes to represent relationships.

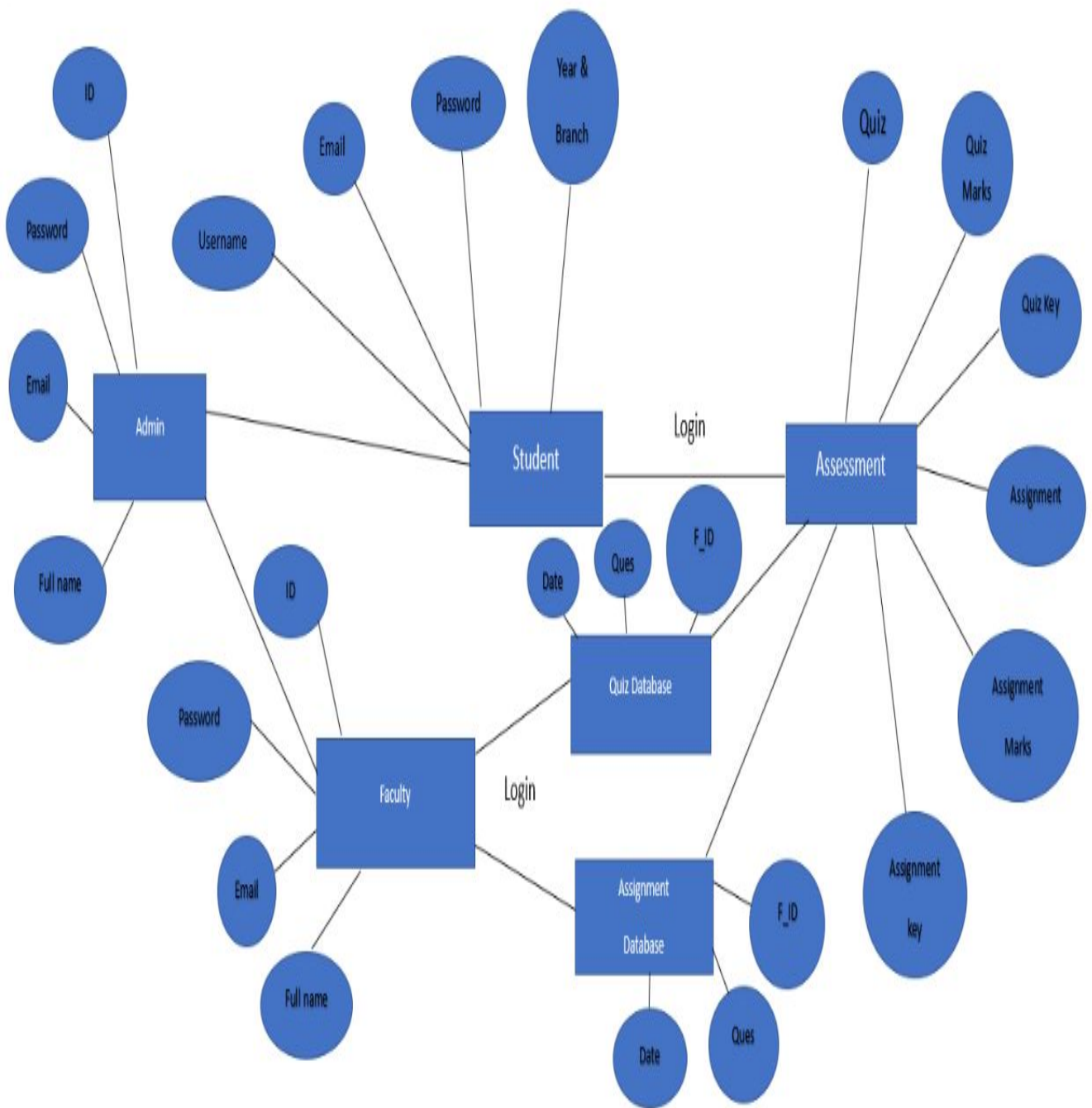


Figure 6.8: Realtion Diagram

6.1.6 Gantt Chart

A Gantt chart is a project management tool that illustrates work completed over a period of time in relation to the time planned for the work. It typically includes two sections: the left side outlines a list of tasks, while the right side has a timeline with schedule bars that visualize work. The Gantt chart can also include the start and end dates of tasks, milestones, dependencies between tasks, and assignees.

In Figure 6.9 we show our work flow of project ,firstly it's shows the planning phase with the hilghlighted lines between months that generally indicates the completion of the phases ,like planning phases all the phases including Research ,Design, Implementation and Follow up were shows the same cases of completion that show's in Gantt Chart with the highlighted lines symbol.

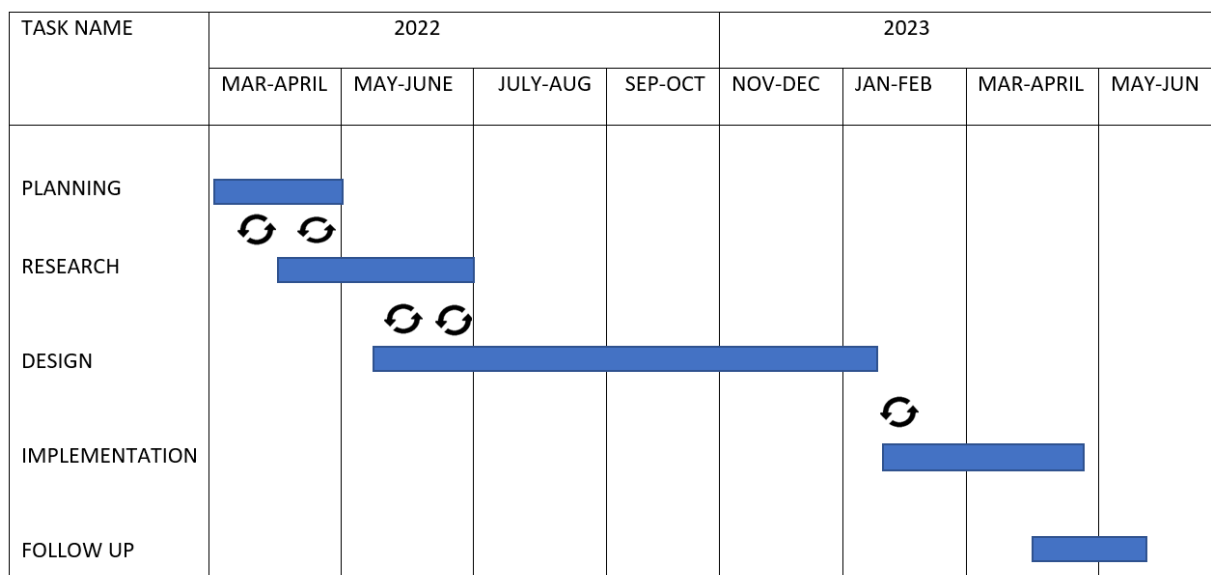


Figure 6.9: Gantt chart

Chapter 7

Implementation

7.1 Proposed Methodology

Algorithm used

- Grammar Checker
- Keyword matching
- Similarity checking

7.1.1 Grammar Checker

A key component of natural language processing is the automatic detection and correction of errors seen in English word written by non-native speakers of the language. This makes use of computer programming technology. Only using computer programming technology to analyse and study these issues will help improve the ability of language processing and recognition and enable natural language processing machines to evolve into machines with human intelligence. This is due to the enormous amount of language vocabulary, the complexity of grammar rules, the ambiguity of semantics, and the ambiguity of speech.

There are currently three methods for correcting grammar errors: the first is rule-based, which formulates specific rules to correct specific error types and heavily relies on the quality of the rules while only being able to modify certain types of errors; the second is statistics-based, which uses related information like words to obtain text features, model the language, and choose appropriate statistical models to correct errors in the text; and the third is model-based,

which corrects errors in the text using mathematical models. In order to correct English grammar in this work, a more precise and effective approach is thus needed.

Sample Code

```
import language_tool_python
def GrammarChecker(solution) :
    my_tool = language_tool_python.LanguageTool('en - US')
    matches = my_tool.check(answer)
    myMistakes = []
    for rules in matches :
        if len(rules.replacements) > 0 :
            myMistakes.append(answer[rules.offset :
            rules.errorLength + rules.offset])
    return len(myMistakes)
```

7.1.2 Keyword Matching

Some grammar checker software have a feature called "keyword matching for marking" that enables teachers and instructors to grade student's writing based on a list of predetermined words or phrases. This function is frequently employed in educational contexts to evaluate tasks, articles, and essays that need for particular vocabulary or language. The keyword matching feature works by analysing the text and identifying instances where specific keywords or phrases are used. Trainers can compile a list of terms or phrases pertinent to the task or subject, then use the grammar checker to locate these words or phrases in the text. By using keyword matching for marking, educators can assess the students' understanding of the material and evaluate how effectively they have incorporated relevant terminology into their writing. This feature also helps ensure that students have met specific requirements or guidelines for the assignment.

Sample Code

```
def KeyWordmatching(X,Ylist) :  
    result = 0  
    Xlist = word_tokenize(X)  
    sw = stopwords.words('english')  
  
    l1=[];l2=[]  
  
    Xset = w for win Xlist if not win sw  
    for Y in Ylist  
        Ylist = word_tokenize(Y)  
        Yset = w for win Ylist if not win sw  
        rvector = Xset.union(Yset)  
        for win rvector :  
  
            if w in Xset : l1.append(1) create a vector  
        else :  
            l1.append(0)  
        if win Yset : l2.append(1) else :  
            l2.append(0) c = 0  
  
        for i in range(length(rvector)) :  
            c+= l1[i]*l2[i]  
        cosine = c / float((sum(l1)*sum(l2))**0.5)  
        cosine = cosine*100  
        result=result+cosine  
  
    cosine=result/3]  
  
    kval = 0 if cosine > 90: kval = 1  
    elif cosine > 80: kval = 2 elif cosine > 60: kval = 3  
    elif cosine > 40: kval = 4  
    elif cosine > 20: kval = 5 else:
```

```
kval = 6  
return kval
```

7.1.3 Length Checking

The Length Check processor provides a quick and easy way for checking an attribute for values of the appropriate length.

The Length Check can check either, or both, of the following: The total length in characters (including whitespace and control characters) The number of words.

A length checker is a tool or function that allows you to determine the length of a given input, such as a string or an array. It is commonly used in programming or data processing tasks to validate the length of input data or to perform specific actions based on the length.

In many programming languages, the length of a string can be obtained using a built-in function or method. For example, in Python, you can use the `len()` function to determine the length of a string:

Sample Code

```
def CheckLenght(clientanswer) :  
    clientans = len(clientanswer.split())  
    return clientans  
    kval1 = 0  
    if clientans > 50 :  
        kval1 = 1  
    elif clientans > 40 :  
        kval1 = 2  
    elif clientans > 30 :  
        kval1 = 3  
    elif clientans > 20 :  
        kval1 = 4  
    elif clientans > 10 :  
        kval1 = 5
```

```
else :  
    kval1 = 6  
return kval1
```

7.1.4 Padding

Sometimes filter does not fit perfectly fit the input image. We have two options:

- Pad the picture with zeros (zero-padding) so that it fits.
- Drop the part of the image where the filter did not fit. This is called valid padding which keeps only valid part of the image.

7.1.5 NLP (Natural Language Processing)

- This is used to extract features.
- NLP is used to analyse text, allowing machines to understand how humans speak. This human-computer interaction enables real-world applications like automatic text summarization, sentiment analysis, topic extraction named entity- recognition parts-of- speech tagging relationship extraction stemming and more.
- NLP in our project is commonly used for automated text mining & machine translation
- NLP is characterized as a difficult problem in computer science. Human language is rarely precise, or plainly spoken.

7.2 Tools and Technologies Used

7.2.1 Tools

- **Python** Python is a high-level, general-purpose programming language. Its design philosophy emphasizes code readability with the use of significant indentation via the off-side rule. Python is dynamically typed and garbage-collected.

- **Vs-Code** Visual Studio Code combines the simplicity of a source code editor with powerful developer tooling, like IntelliSense code completion and debugging. First and foremost, it is an editor that gets out of your way. The delightfully frictionless edit-build-debug cycle means less time fiddling with your environment, and more time executing on your ideas.

7.2.2 Technologies

- **Node Js** Node.js is a server-side platform built on Google Chrome's JavaScript Engine (V8 Engine). Node.js was developed by Ryan Dahl in 2009 and its latest version is v0.10.36. The definition of Node.js is a platform built on Chrome's JavaScript runtime for easily building fast and scalable network applications. Node.js uses an event-driven, non-blocking I/O model that makes it lightweight and efficient, perfect for data-intensive real-time applications that run across distributed devices.

Node.js is an open source, cross-platform runtime environment for developing server-side and networking applications. Node.js applications are written in JavaScript, and can be run within the Node.js runtime on OS X, Microsoft Windows, and Linux.

Node.js also provides a rich library of various JavaScript modules which simplifies the development of web applications using Node.js to a great extent.

- **MONGO DB** The most well-known NoSQL database, MongoDB, is a free, document-oriented database. Non-relational is what "NoSQL" refers to. It means that MongoDB offers a completely alternative mechanism for data storage and retrieval and is not based on the relational database structure that resembles a table. The name of this storage format is BSON. Tabular data is stored in SQL databases. This data is kept in a preset data model that isn't very flexible for the modern, rapidly expanding real-world applications. Applications today are more social, interactive, and networked than ever before. Applications are storing and accessing data at increasing and higher speeds. Due to their architecture, relational database management systems (RDBMS) are not the best option for handling massive data because they are not horizontally scalable. The database will hit a scaling limit if it uses a single server. NoSQL databases offer better performance and are more scalable. With its adaptable document model, MongoDB is a NoSQL database that scales by adding more and more servers while boosting efficiency.

- **Android Studio** Android just overtook Windows as the most widely used operating system in the world for a number of reasons. I want to explain what the Android Studio is as an Android developer. The Google Android Development IDE, Android Studio, was introduced on May 16, 2013, during the Google I/O 2013 conference. All the Android tools you need to create, test, debug, and profile your application are included in Android Studio. The build automation tool Gradle is used by the Android Studio to manage your project. You must download Android Studio for your choice operating system (Windows®, Mac OS X, or Linux) from the Android developers website in order to create your first app. Your application may be created and tested in Android Studio on either a real device or an emulator.
- **Jupyter Notebook** To create and share documents with live code, equations, visualisations, and text, you can use the free and open-source Jupyter Notebook web application. The staff of Project Jupyter is in charge of maintaining Jupyter Notebook. The IPython project, which formerly had an IPython Notebook project of its own, gave rise to Jupyter Notebooks. The primary programming languages it supports are Julia, Python, and R, hence the name Jupyter. There are presently more than 100 additional kernels available, however Jupyter comes with the IPython kernel, which enables Python programming.

7.3 Module Used

7.3.1 Preprocessing Module

Once the user enters their inputs, the solution and the response are both preprocessed. Some preprocessing methods include tokenization, stemming, stop word removal, case folding, and finding and inserting synonyms word into the normal word(text). Because word2vec has a large dictionary and can make use of such stop words to improve the text's semantic value, it is important to note that stop text are eliminated not when the input is passed to it. Stop words are eliminated before being transmitted to a machine learning model like Multinomial Bayes, despite the fact that they make it more difficult for the computer to recognise patterns.

7.3.2 Result predicting module

Centrepiece of study is the Result Predicting Module. Figure 3 depicts how this module operates. It uses the first algorithm listed below: Now that we have the maximum matched solution/answer pair scores, we can calculate the overall score using either WDM or Cosine Similarity. This outcome can be used to calibrate a machine learning model or compared to a real score, is selected.

- To forecast the class or category of the preprocessed answer, a machine learning model uses testing data, and the category is verified using the output of the result prediction module. This increases our level of confidence in the model's predicted outcome.
- The machine learning model receives the preprocessed response and its label. In addition, the model is updated in light of recent data. The final grade, the answer, the response, and the projected class are all sent to the module that predicts the ultimate score. The model is initially fairly sparse and inaccurate, thus it is best to train it with a set quantity of data before beginning to test it.
- The last score, the response, the solution, and the anticipated class are given to the last score projection. It is recommended to train the model with a preset set of data before examining it because of the model's initial sparseness and fuzziness. The benefit of the model serving as a confidence builder for the solution Prediction Module when adequately data-tested. It can be used independently to forecast grades or classes for replies once it has been properly trained on a sufficient amount of data. The word2vec and score Prediction Module that were previously described become obsolete as a result of the creation of a model that can act as an objective assessor for that particular task. Furthermore, it helps with handling the unusual circumstances where the word2vec model, which is less the outcome for a particular answer due to semantic dissimilarity model.

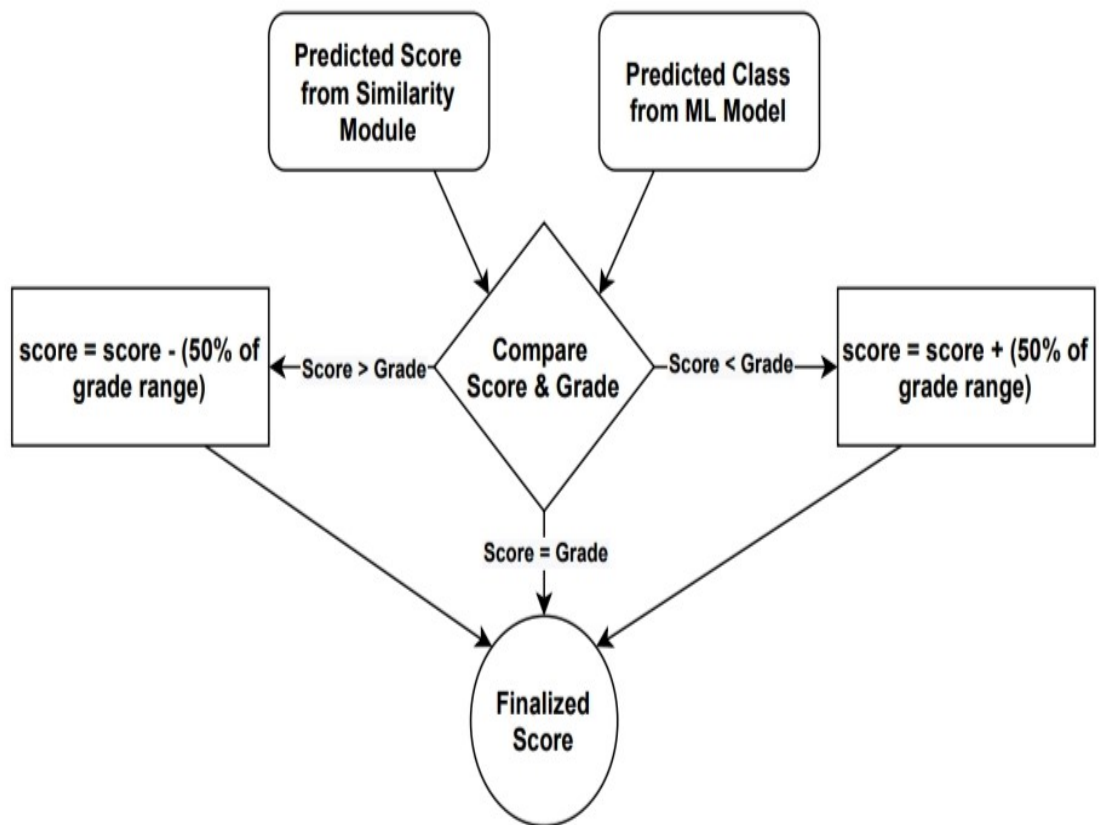


Figure 7.1: Result Prediction Module

Chapter 8

Testing

8.1 Accuracy Tale

Sr.no	Actual Score	Predicted Score	Accuracy
1.	20/20	20/20	100%
2.	6/20	4/20	60%
3.	10/20	8/20	80%

Table 8.1: Accuracy Table

Overall Accuracy- 82%

Description: Test no. is the serial number of tests given by our college ,and the actual score is the score gien by the faculty member which we compare with our module predicted score and we get overall 82% accuracy.

8.2 Test Cases Table

8.2.1 Signup Positive

Here we take dummy credential for the testing phase and check the user if user not exist then this system add new user in the database. In this figure 9.1: we show the test case scenario all the inputs were match the Expected Output with the Actual output if all Criteria matched then access granted for the user.

Test Scenario -ID	Signup -1	Test case -ID	Signup- 1A			
Test case Description	Signup-Positive	Test Priority	High			
Pre-requisite	A valid user account	Post-requisite	NA			
Test Execution Steps:						
S.no.	Actions	Inputs	Expected Output	Actual Output		Test Result
1.	Fill all the fields of registration Page	Email: amanojha434@gmail.com Username:aman Password: **** Confirm Password : ****	Access Granted	Access Granted		PASS
2.	Click on the Register Button	Click on the Register Button	Data Store in the Database	Data Store in Database		PASS

Figure 8.1: Positive Sign-up

8.2.2 Signup Negative

Here we take dummy credential for the testing phase and check the user if user already exist or password mismatch then this system don't add user in the database. In this figure 9.2: we show the test case scnenario ,if any password mismatch or issuse with the username then access denied for the further action.

Test Scenario -ID	Signup -2	Test case -ID	Signup- 2A		
Test case Description	Signup-Negative	Test Priority	High		
Pre-requisite	A invalid user account	Post-requisite	NA		
Test Execution Steps:					
S.no.	Actions	Inputs	Expected Output	Actual Output	Test Result
1.	Fill all the fields of registration Page	Email: alfahad434@gmail.com Username:aftab Password: ***** Confirm Password : **** (Invalid Password)	Access Denied & Application display Error Message	Access Denied & Application display Error Message	PASS
2.	Click on the Register Button	Click on the Register Button	Data not Store in the Database	Data not Store in Database	PASS

Figure 8.2: Negative Sign-up

Chapter 9

Final Interface

9.1 Login Interface

' In figure 10.1: page shows the login interface of our project which will enable user to access our project. The criteria to login is the student has to give his username and password then if his credentials is match then he will be redirected to the quiz page.

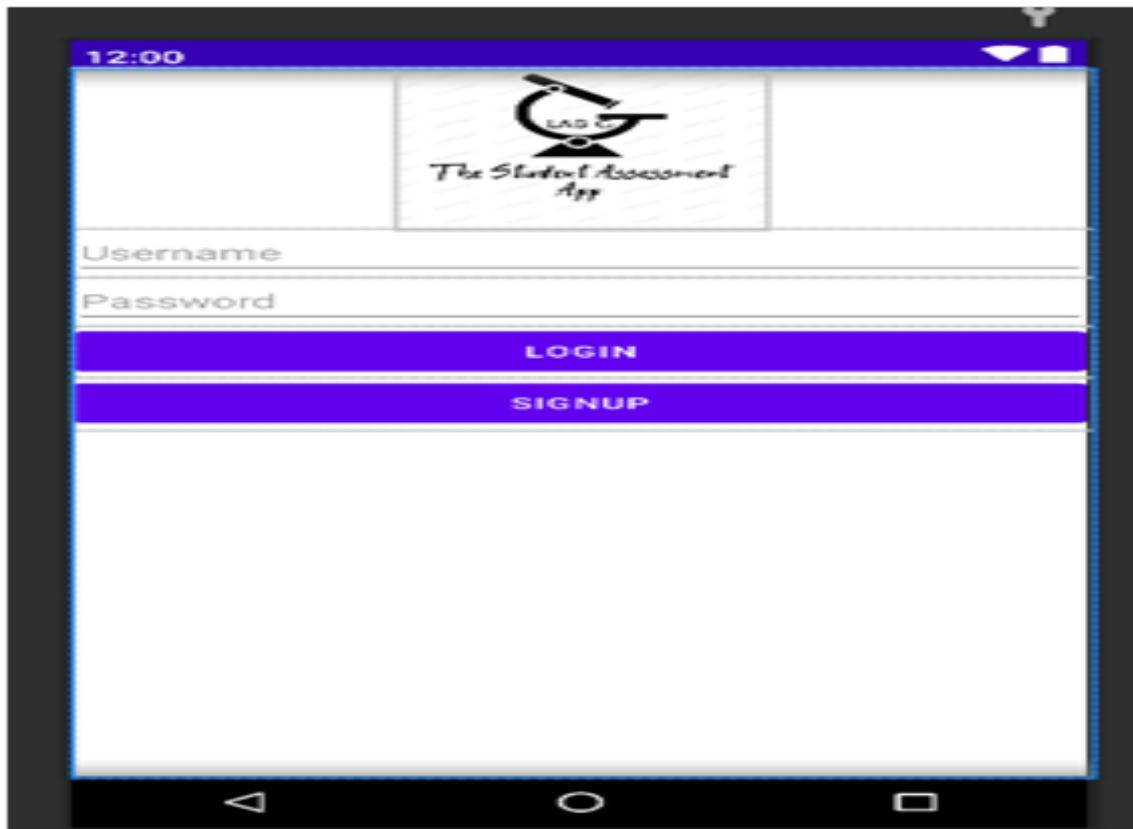


Figure 9.1: Login Interface

9.2 Register Interface

In Figure 10.2: shows the register interface of our project in which students were to fill all the credentials which they asked for completing his registration for next follow up actions in the dashboard.



The screenshot displays a mobile application interface for registration. At the top, a status bar shows the time as 12:00. Below this, a header section features a logo with a microscope and the text "The Student Assessment App". The main form consists of several input fields: Username, Email, Password, Student_ID, Collage_ID, Batch, Semester, and Branch. A prominent red button labeled "REGISTER" is positioned at the bottom of the form. The interface is set against a light blue background with a subtle grid pattern.

Figure 9.2: Register Interface

9.3 Student Assignment Interface

In this figure 10.3: In this figure , we show the interface of assignment page for student where they retrieve assignment from there faculty and write down there answer after .

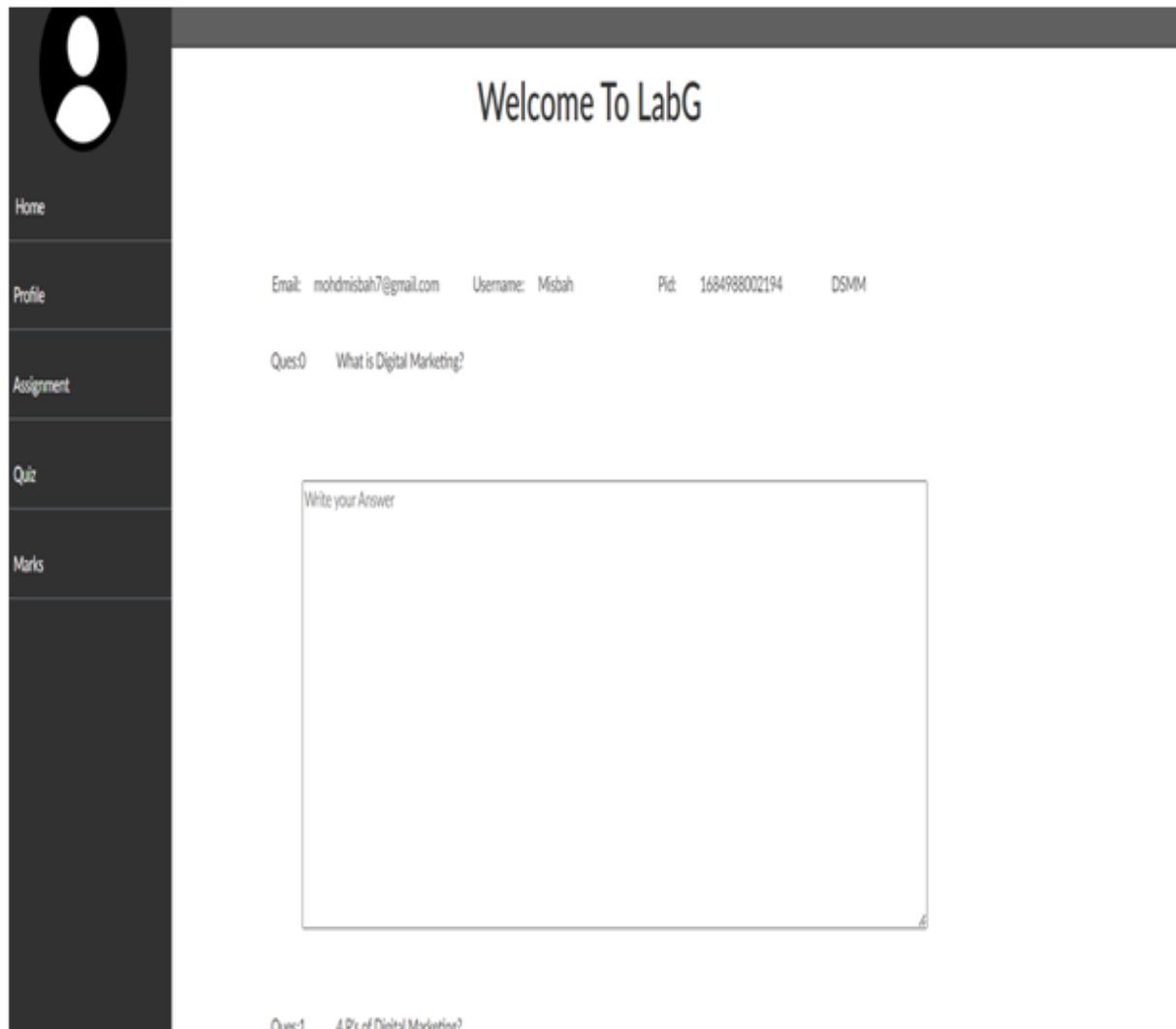


Figure 9.3: Student Dashoard Interface

9.4 Faculty Interface

In this figure 10.4: we show the interface of assignment page for faculty where they can create assignment by providing number of questions with there general answer which use for evaluation of student answer for the assignment.



The image shows a web interface for a faculty dashboard. On the left is a dark sidebar with a circular profile icon at the top. Below the icon are five menu items: 'Home', 'Profile', 'Assignment', 'Quiz', and 'Marks'. The main content area has a dark header with the text 'Welcome To LabG'. Below the header, there is a row of three fields: 'Email: mohdmiab454@gmail.com', 'Pid: 1684968002194', and 'DSMM'. The central part of the page contains two text input fields. The first field is labeled 'Why Digital marketing needed?' and contains the text 'Because of beneficial promotion and digital evolution.'. The second field is empty. At the bottom of the main content area, there are two buttons: 'Submit' and 'Not'.

Figure 9.4: Faculty Dashboard Interface

Chapter 10

Conclusion and Future Work

10.1 Conclusion

When employed with examine data from the equal data set used to train the classifiers . The assuming strategy performs the assumption of student's lab performance. But the predictor consistently misidentifies the contempt related penalty.This is because there aren't many clear examples of contempt in the field of lab sessions.This applications resolves the problem of Assessments in collage which has two functions first is evaluation of the student on the basis of their online assessment and second is the shows of student's weak areas . With each evaluation, it helps the students demonstrate their understanding, provides feedback for their errors, and offers opportunities for progress.It totally ensures the problem of information security with the help of hashing . It also create a virtual environment for student to give test with a more interactive way with time limit questions.

10.2 Future Scope

In schools, assessments serve two goals. Each exam provides an opportunity for the students to demonstrate what they have learned, get feedback on their errors, and enhance their performance. Teachers can use it as a fantastic tool to evaluate the efficacy of their preferred method of instruction. What does assessment mean in terms of education? Information is gathered from a variety of sources during an assessment to determine the student's level of understanding and expertise.

Appendix A

Conceptual Background

As was previously stated, the idea of assessing subjective responses has been researched for around 20 years. The Bayes theorem, Knearest classifier, large data natural language processing, comparable checker, and even formal techniques like formal concept analysis have all been used to solve this problem. Statistical, information extraction, and full natural language processing are the three main categories into which they can be divided.

Statistical Technique: It is based on keyword matching and is seen as being ineffective because it can't handle issues like synonyms or take context into consideration. This method has been used in several works on subjective paper evaluation.

Information Extraction (IE) Technique: Information Extraction techniques depend on getting a structure or a pattern from the text so that the text can be broken into concepts and their relationships [11]. The dependencies found to play a significant role in producing scores and need to be confirmed from an expert in domain [12], [13].

Natural Language Processing(NLP): These methods parse the text and determine its semantic meaning utilising tools for natural language processing [14], [15]. To determine the final score, that meaning can then be contrasted with the meaning discovered from the solution. Preprocessing is the process of preparing text documents for machine processing. It involves a number of natural language processing techniques, including tokenization, stopword removal, parts of speech tagging, lemmatization, Automated scoring systems and their integration with machine learning and natural language processing were covered to calculate tree similarity.

Bibliography

1. Armstrong, M., Clarkson, P. and Noble, M. (1998) Modularity and credit frameworks: the NUCCAT survey and 1998 conference report, Newcastle-upon-Tyne, Northern Universities Consortium for Credit Accumulation and Transfer.
2. Bridges, P., Cooper, A., Evanson, P., Haines, C., Jenkins, D., Scurry, D., Woolf, H. and Yorke, M (2002) 'Coursework marks high examination marks low: discuss', *Assessment and Evaluation in Higher Education*, vol. 27 (1),
3. Brown, S., Rust, C. Gibbs, G (1994) *Strategies for diversifying assessment in higher education*, Oxford, Oxford Centre for Staff Development
4. Dalziel, J (1998) 'Using marks to assess student performance: some problems and alternatives', *Assessment and Evaluation in Higher Education* 23 (4), pp 351-366
5. Elander, J. and Hardman, D. (2002) 'An application of judgement analysis to examination marking in psychology', *British Journal of Psychology* 93, pp. 303-328.
6. Falchikov, N. (2004) *Improving Assessment Through Student Involvement: Practical Solutions for Higher and Further Education Teaching and Learning*, London: Routledge
- Falmer Forbes, D., Spence, J. (1991) 'An experiment in assessment for a large class'. In R. Smith (Ed.), *Innovations in engineering education*. London: Ellis Horwood.
7. Gibbs, G. Simpson, C. (2002) Does your assessment support your students' learning available at: <http://isis.ku.dk/kurser/blob.aspx?feltid=157744>
8. House of Commons Innovation, Universities, Science and Skills committee (2009) *Students and universities*, London: The Stationary Office Ltd.
9. Kember, D., et al. (1997) 'Case studies in improving teaching and learning from the action learning project', *Action learning project*, Hong Kong.

10. King D (2008) 'Does it make a difference? Replacing text with audio feedback'. *Practice and Evidence of the Scholarship of Teaching and Learning in Higher Education* vol 3 issue 2 pp 145-163
11. Laming, D. (1990) 'The reliability of a certain university examination compared with the precision of absolute judgments', *Quarterly Journal of Experimental Psychology*, vol. 42 pp. 239-254
12. Rust, C., O'Donovan, B., Price, M. (2005) 'A social constructivist assessment process model: how the research literature shows us this could be best practice', *Assessment and Evaluation in Higher Education*, Vol. 30 (3)
13. Sadler, D. R. (1987), 'Specifying and Promulgating Achievement Standards', *Oxford Review of Education*, 13, pp. 191-209.
14. Yorke, M. (1997) 'Module mark distribution in eight subject areas and some issues they raise', in N. Jackson (Ed), *Modular higher education in the UK*, London: Higher Education Quality Council, 105-107
15. Zhang, L. F. and Watkins, D. (2001), 'Cognitive development and student approaches to learning: an investigation of Perry's theory with Chinese and US university students', *Higher Education*, 41, pp. 236-261.